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In situ fluorimetry: A powerful non-invasive diagnostic technique for natural dyes used in artefacts
Part I. Spectral characterization of orcein in solution, on silk and wool laboratory-standards and a fragment of Renaissance tapestry

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Abstract

In this paper the potentialities of spectrophotometric and fluorimetric techniques for identifying the materials used in artistic textiles are investigated. A portable non-destructive instrument suitable to record fluorescence spectra on surfaces was set up and successfully experienced. A naturally occurring dye, orcein, which was widely used in antiquity for textile dyeing, has been spectrally characterized in both solution and powder. Laboratory samples of wool and silk orcein-dyed threads were analysed before and after ageing. An original fragment of Renaissance tapestry was also analysed. The textile (wool) and the colourant (orcein) were recognised by comparison with the data from the laboratory samples.
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Keywords: Fluorimetry; Orcein; Photodegradation; Textile; Tapestry

1. Introduction

Interest in the study of works of art, for the purposes of conservation and restoration, has been increasing in the last years, also stimulated by the technological development which has made available improved diagnostic techniques and especially easily portable instrumentation for in situ non-destructive investigations. While numerous non-destructive analytical techniques have been successfully employed, also in situ, to characterize pigments, including X-ray fluorescence and infrared, Raman and UV–vis spectroscopy, a few tools, and all destructive, are available for the analysis of colourants (namely, reversed-phase HPLC [1], laser desorption mass spectrometry [2], thermally assisted hydrolysis and methylation [3]). As a matter of fact, currently there is no reliable technique for the non-invasive identification of lakes and dyes. Absorption spectrophotometry was proposed for identifying organic colourants [4] and fluorimetric techniques have been experienced to characterize both dyes [5,6] and binders [5]. None of these techniques has been widely applied also because they required sampling and consequently some damage to the artefact. Recently, Clarke has proposed 3-D fluorescence reflectance spectroscopy as a new technique for the non-destructive identification in situ of organic pigments, dyes and inks on early mediaeval manuscripts [7]. However he has also found that fluorescence spectroscopy has some limitations as a tool for non-destructive in situ identification of colourants [8]. Three-dimensional fluorescence was successfully employed for the non-destructive determination of dyes from madder on silk fiber [9].

Up until the 1800th, colourants for dying textiles were extracted from insects or plants [10–12]. In particular, the naturally occurring organic dye, orcein, studied in this paper, was extracted from lichens [13] and has been occasionally used to produce false shellfish, that is to substitute the more precious Tyrian purple. In a historical review on orchil dyes, Kok [14] reports the early use of lichen purple in combination with and as a substitute for shellfish purple to be mentioned by Theophrastus, Pliny the Elder and in the Stockholm Papyrus. The expansion of the use continued in the 16th and 17th centuries.

Natural orcein is a mixture of different reddish dyes, which can give intramolecular-hydrogen bonding. Depending on the pH and the presence of mordants, it can appear red, purple or

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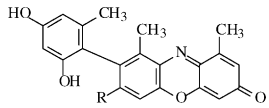
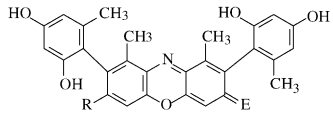
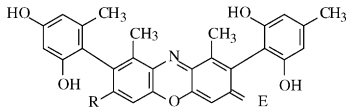


α-amino orcein R=NH_2
α-hydroxy orcein R=OH



β-amino orcein R=NH_2, E=O
β-hydroxy orcein R=OH, E=O
β-amino orcein imine R=NH_2, E=NH



γ-ammino orcein R=NH_2, E=O
γ- hydroxy orcein R=OH, E=O
γ- amino orcein imine R=NH_2, E=NH

Scheme 1.

orange [15,16]. From the dye extracted from lichens, by using high performance liquid chromatography, Musso et al. [17] succeeded in separating 14 compounds and used the synthesis [18] and UV–vis and IR spectroscopy [19] to support the proposed structures of the individual components. The principal eight ones were identified as *trans*- and *cis*-isomers of α-, β- and γ-amino hydroxy-orcein (Scheme 1).

In the present work, attention has been particularly directed at studying the spectral properties of natural orcein and investigating the photochemical and thermal processes, which are responsible for its degradation. Dye solutions, as well as dyed textiles, were investigated. The study was first addressed to spectrally characterize the dye in solution and the solid powdery phase. Then, laboratory samples mimicking ancient textiles were analysed before and after ageing in order to have standards to which the emissions from original and restoring materials could be compared. An original fragment of Renaissance tapestry from Bruxelles (XVI century) was also analysed with the aim to identify both the textile (wool) and the colourant (orcein) by comparison with the laboratory samples.

Reflectance techniques were used in both absorption and emission. To avoid destructive or even microdestructive analyses, a portable fluorimeter was assembled and its performances were checked by comparison with the results obtained using a commercial laboratory instrument. It was tested on materials constituting ancient textiles in the perspective of three basic applications: (i) diagnosis and mapping of original materials; (ii) monitoring during the cleaning procedures to control their efficiency without altering the original material and, (iii) monitoring of the degradation of the original and restoring materials. The fluorescence technique is very sensitive and specific. The portable instrument can be used in a museum as well as on scaffoldings in a restoration yard of wall paintings.

2. Experimental

2.1. Materials

Natural and synthetic orcein were purchased from Acros; a synthetic sample was also obtained from Fluka. Ethanol and acetonitrile were from Merck.

Dyed samples of wool and silk fibres were prepared by dissolving the colourant in water, in both the presence and absence of the mordant (alum), at 40 °C; then the temperature was raised to the boiling point and maintained in these conditions for 20 min. A surfactant (Sandopan) was added to the solution in order to favour dying. Once extracted from the bath, the fibre was washed with water and dried.

2.2. Equipment

Absorption spectra in solution were recorded using a Perkin Elmer Lambda 16 spectrophotometer. For measurements on solid samples, an integrating sphere accessory for reflectance measurements, was used. Emission spectra were obtained using a Spex Fluorolog-2 1680/1 spectrofluorimeter, controlled by the Spex DM 3000F spectroscopy software. To obtain emission spectra from surfaces external to the sample holder of the fluorimeter, measurements were also carried out using a fibre optic sample accessory (Fluorolog-2). By this accessory, the excitation light is directed to the sample via a randomized bifurcated fibre optic bundle and the fluorescence from the sample is redirected to the recto angle port in the sample compartment.

The portable fluorimeter was built-up by assembling the separate components. The exciting light source (75 W xenon lamp, 240–600 nm emission), is focused on a H-10 Jobin Yvon UV-monochromator (linear dispersion 8 nm/mm) for selecting the excitation wavelength. A fibre-optic cable directs the exciting light on the sample. The emitted light is convoyed by another fibre-optic cable to a high sensitivity (86 photons/counting) Avantes CCD spectrometer. The instrument is interfaced to a portable computer with suitable software to store and elaborate the data. To optimise and reproduce the position of the fibre on the sample, an accessory, built up by transparent material, is used which allows the underlying surface to be seen and the fibre position to be carefully selected. The assembled apparatus approximately weights 25 kg and can laid on a mobile table (60 × 60 cm), thus allowing to be easily transported and used even in small rooms. This equipment for in situ reflectance measurements of UV–vis absorption and fluorescence has suitable sensitivity and resolving power.

For the accelerated photo-ageing trials, a 150 W xenon lamp, filtered through a pyrex filter, was used.

3. Results

3.1. Spectral characterization of natural orcein

The absorption spectrum of the natural dye in acetonitrile (Fig. 1) exhibits a maximum at 507 nm, which is markedly shifted to the red in ethanol (λ_{max} = 567 nm). The behaviour

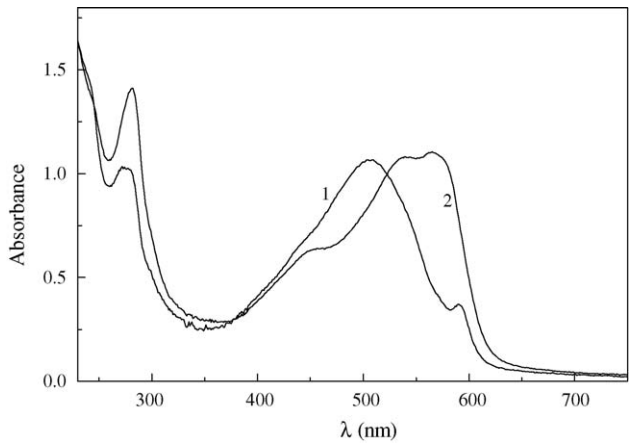


Fig. 1. Absorption spectrum of natural orcein in acetonitrile (1) and ethanol (2).

in the latter solvent has been assigned to formation of solute–solvent molecular complexes where the solvent plays the role of a proton acceptor [15]. From the absorption spectra monitored as a function of the pH (see below), it can be seen that there is a close similarity with the spectrum in an alkaline solution, where anionic dye species are surely present.

The emission and emission excitation spectra of orcein in acetonitrile are quite specular (Stokes shift, $\Delta\nu = 2360\,\mathrm{cm}^{-1}$) and the excitation spectrum well corresponds to the absorption spectrum (Fig. 2). In ethanol, a more complex behaviour was observed (Fig. 3). Excitation at 520 nm produces a narrow band at 595 nm, whereas a broader band is monitored when exciting at 420 nm, which is probably due to overlap of a new emission with the previous one. This behaviour suggests the presence of at least two fluorophores. The red shifted excitation could be due to some deprotonated forms (see below).

Since tapestries are exposed to deterioration because of ageing and also due to atmospheric pollutants, it is important to know which chromatic changes they undergo when the environmental conditions change. During the last decades, acidic rains have been one of the most important causes of pollution. For this reason, we investigated the effect of pH on the absorption and fluorescence spectra of orcein in aqueous solution. Since we are dealing with a mixture of compounds, the

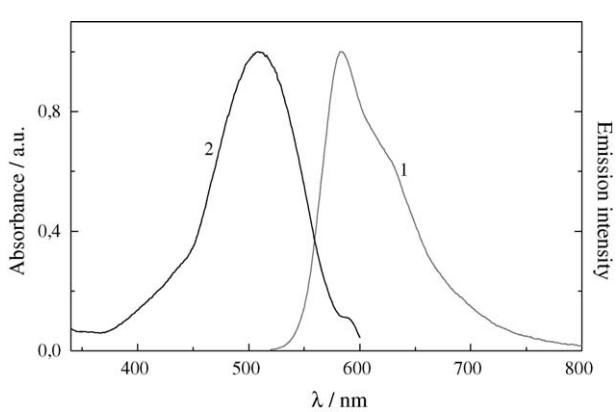


Fig. 2. Normalised emission (1) and emission excitation (2) spectra of natural orcein in acetonitrile.

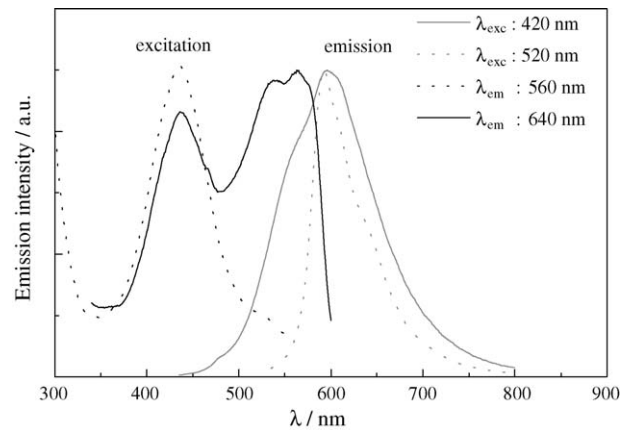


Fig. 3. Normalised emission (grey) and emission excitation (black) spectra of natural orcein in ethanol.

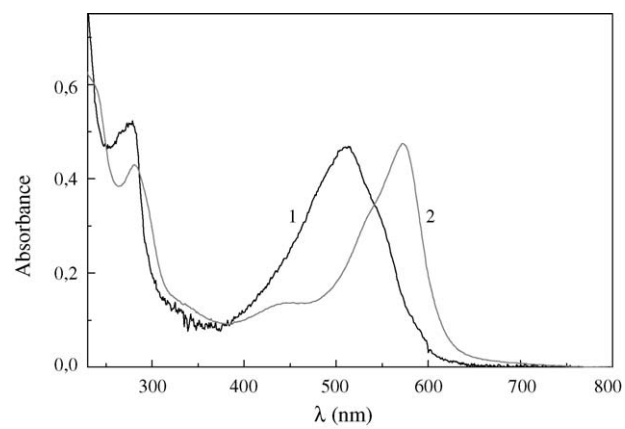


Fig. 4. Effect of the pH on the absorption spectra normalised of orcein in aqueous solutions: (1) pH 2 and (2) pH 13.

changes observed cannot be attributed to individual acid–base equilibria, however, they are meaningful since they indicate that the dye exhibits an acidichromic behaviour. In fact, the absorption maximum observed at 510 nm in acidic solution, shifts to 570 nm in a basic solution and the colour turns from purple to blue-purple (Fig. 4). The spectral red-shift on going from acidic to neutral and basic solutions is a common feature of several dyes containing hydroxyl groups, due to OH deprotonation [20–22].

The fluorescence emission of orcein in aqueous solution showed a maximum at 595 nm, independent of the pH. In powdery solid phase, the emission maximum was markedly shifted with respect to the solution (645 nm). This behaviour was also observed for other organic dyes [23].

3.2. Spectral characteristics of the fibre standards

To analyse the chromatic effects due to the interactions of the dye with the textile and mordant, five standard samples were prepared for both wool and silk fibres. In each series, one sample was uncoloured while the others were dyed with natural and synthetic orcein (for the purpose of comparison) in the absence and presence of a mordant (alum). The use of mordants generally improves the light-fastness [24]. The five silk

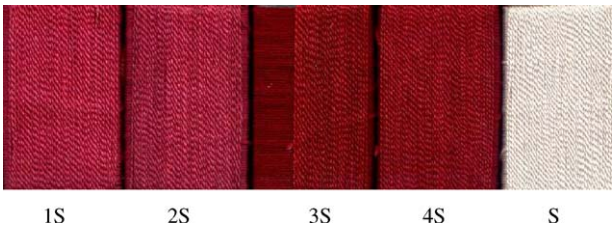


Fig. 5. Silk standards: 1S, natural orcein-dyed silk; 2S, natural orcein and alum-dyed silk; 3S synthetic orcein-dyed silk; 4S, synthetic orcein- and alum-dyed silk; S, untreated silk.

samples are shown in Fig. 5; similar samples were prepared with wool.

The reflectance absorption spectra of the uncoloured silk and wool samples exhibited a broad band at about 300 nm, whereas the value reported in the literature for wool is at somewhat higher energy (250–290 nm) [25]. This discrepancy can be due to the presence of a greater amount of photodecomposition products present in our samples. Both the fibre and the colourant absorptions were observed on the dyed samples. The absorption spectra of natural orcein on silk and wool were detected in the same wavelength range (520–580 nm) and were quite similar to the spectrum in ethanol (λ_{max} = 520–567 nm). This suggests that some of the dye molecules act as proton donors towards the fibre. The absorption patterns of the samples dyed with synthetic orcein extend towards longer wavelengths, thus resulting darker in colour (samples 3S and 4S in Fig. 5). Chromatic differences between samples with and without mordant could not be detected.

Freshly prepared standards were excited using 254 nm light, which is absorbed by both the fibre and the dye. Two emission bands were observed, one around 450 nm and the other, more intense, at about 640–650 nm (Fig. 6). Comparison of the emissions from coloured and colourless samples allowed the ipsochromic emission to be attributed to the fibre and the bathochromic one to the colourant. No difference in fluorescence was detected between samples dyed in the absence and in the presence of alum.

In Table 1, the emission maxima monitored in different media are collected. It can be seen that the emissions detected on textiles occur at wavelengths close to that monitored in the solid state. Both show a marked bathochromic shift compared to solution.

3.3. Ageing of the samples

Due to natural ageing and exposure to solar irradiation, deterioration of the dyed textiles, associated with changes in the original colour, may occur. In order to evaluate orcein deterioration, dyed and not dyed samples were exposed to natural and artificial ageing and subsequently analysed.

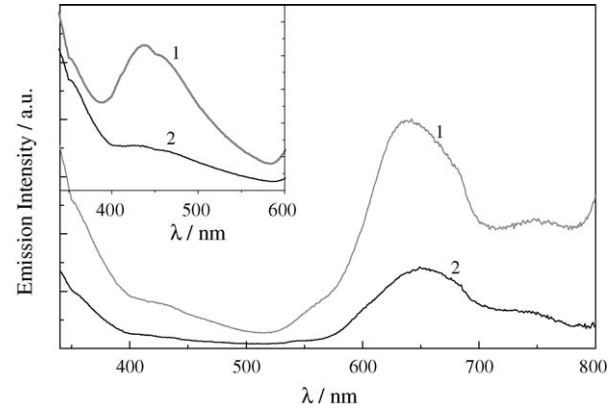


Fig. 6. Emission spectra of silk (1) and wool (2) textiles freshly dyed with orcein. Inset: emissions from the untreated silk (1) and wool (2) samples.

The time evolution of an uncoloured silk sample exposed to artificial ageing by irradiating with a xenon lamp is shown in Fig. 7. The emission band, with the maximum around 450 nm, showed a fast decrease during the initial irradiation times and reached an almost constant value after about 20 irradiation hours. A similar behaviour was found for wool samples. This behaviour is in contrast with the literature finding that the 450 nm emission, which was assigned to degradation products from wool, increased as the exposure time increased [26].

Accelerated ageing of both orcein-dyed silk and wool samples caused an increase in the emission intensity of the textile

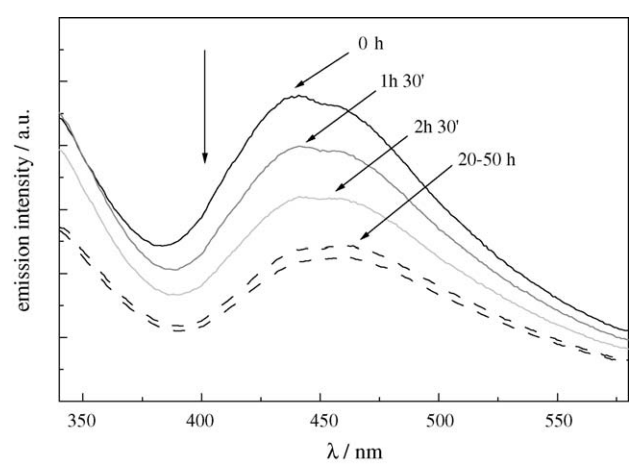


Fig. 7. Time course of the emission of the silk uncoloured standard (λ_{exc} = 254 nm) under irradiation with a filtered xenon lamp.

Table 1
Emission maxima (shoulders) of natural orcein in solution, textiles and solid phase

EtOH	CH_3CN	Acidic pH	Neutral pH	Basic pH	Silk	Wool	Powder
595 (543)	580 (625)	595	595	595	640 (750)	650	645

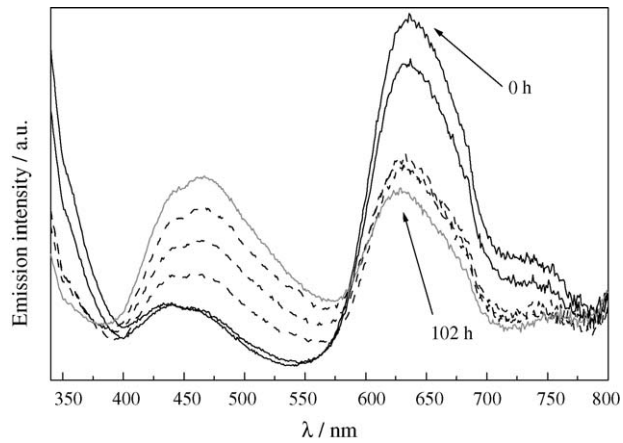


Fig. 8. Emission from the sample of natural orcein on silk before ($t = 0$) and upon accelerated ageing.

and a parallel decrease in that of the dye, as shown in Fig. 8 for an orcein-dyed wool sample. This change was accompanied by a blue shift of the maximum of the colour band, as shown in Fig. 9 for orcein on silk, where the spectra recorded at various irradiation times are normalised.

The increase of the textile signal, which seems to be in contrast with what was observed for the not-dyed sample, can be explained considering two effects: (i) the incident light that the thread absorbs is filtered by the colourant and (ii) the light emitted by the thread is reabsorbed by the dye. Both effects contribute to reduce the emission from the thread, but both decrease as the degradation of the dye proceeds. The ratios of the emission intensity of the textile to that of orcein are plotted as a function of ageing time in Fig. 10 for both the wool and silk samples. As can be seen from the figure, the decolouration of the dye is faster on silk than on wool.

The colour bleaching was also well evidenced by the reflectance spectra monitored before and after irradiation. The time course under irradiation of the absorption spectrum of an orcein-dyed silk sample is shown in Fig. 11 in terms of pseudo-absorbance, $A' = \log\ (1/R)$ [27]. The band

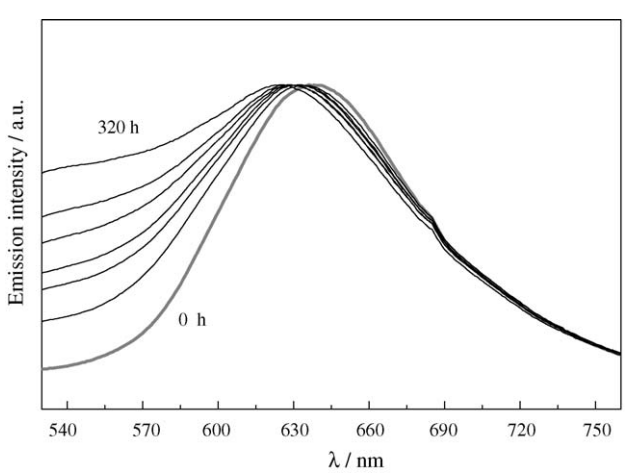


Fig. 9. Time course of the emission spectrum of orcein on wool under 440 nm light excitation. The spectra are normalised on the maximum.

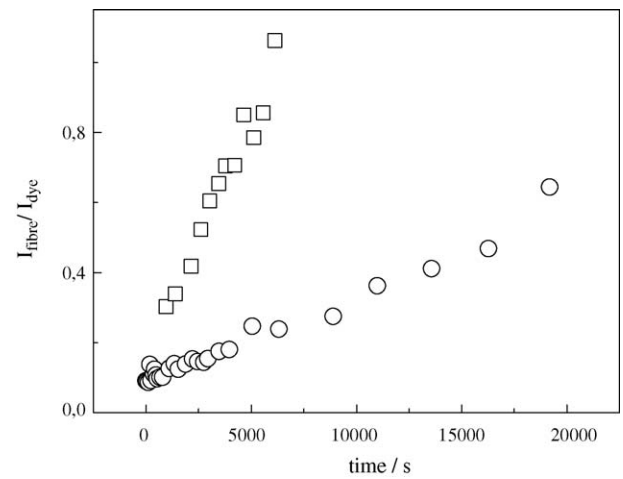


Fig. 10. Ratios of the emission intensity of the fibre to that of orcein plotted as a function of ageing time. Squares refer to silk and circles to wood.

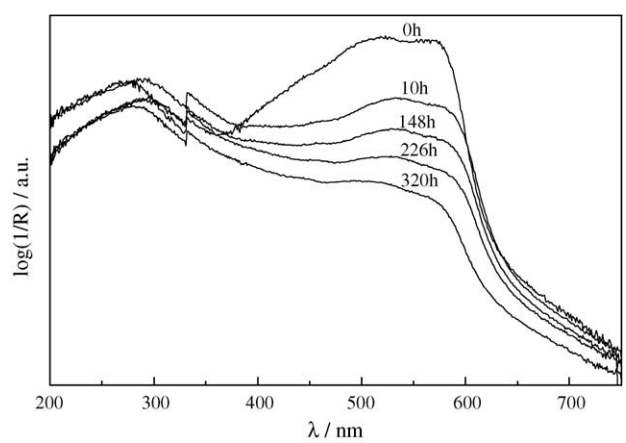


Fig. 11. Time course under irradiation of the absorption spectrum of orcein on silk.

at shorter wavelengths (280 nm) can be assigned to the fibre [25].

The dyed samples were also naturally aged, keeping them exposed to daylight at ambient temperature over a period of 10

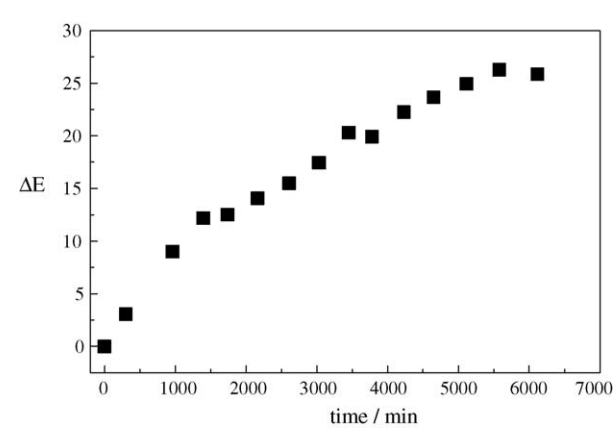


Fig. 12. Colourimetric ΔE as a function of the irradiation time for an orcein-dyed silk sample.

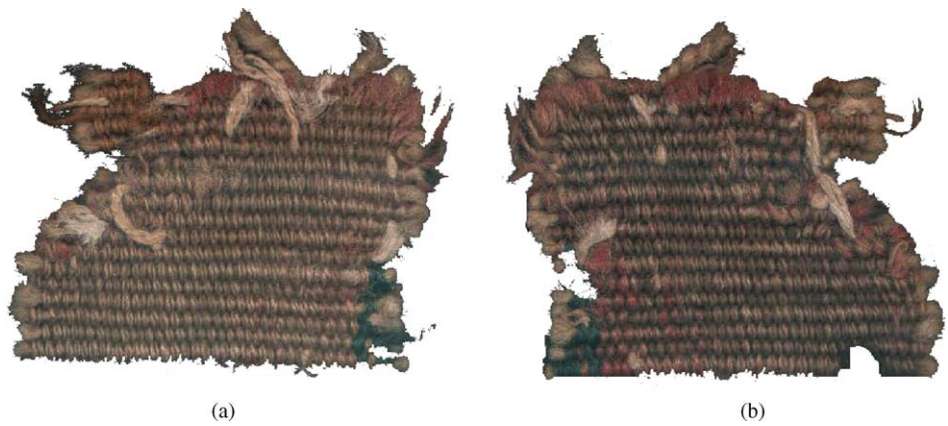


Fig. 13. Fragment of tapestry from Bruxelles (XVI century): (a) front and (b) back.

Table 2
Variations in the three-chromatic coordinates of orcein-dyed wool and silk samples, after an equal irradiation time (100 h)

	Wool sample	Silk sample
ΔL^*	9	16
Δa^*	−8	−18
Δb^*	4	9
ΔE	13	26

months (from August to June of the next year). After ageing, the colour bleaching could also be detected by eye. The values of the L^*, a^*, and b^* colourimetric three-chromatic coordinates were determined before and after ageing of the wool and silk samples; ΔL^*, Δa^*, and Δb^* are reported in Table 2. Their values indicate an increase in luminance and yellow component and a decrease in the red component.

The parameters describing the global effect $[\Delta E = (\Delta L^{*2} + \Delta a^{*2} + \Delta b^{*2})^{1/2}]$ after irradiating for 100 h are also shown in the table. ΔE is plotted as a function of the irradiation time in Fig. 12 for an orcein-dyed silk sample. The behaviour of wool samples was qualitatively similar. However, as observed for other dyes, even orcein was more photodegradable on silk [28]. To obtain a comparable effect on silk and wool, the exposure had to be doubled for the latter.

3.4. Spectrofluorimetric analysis of a Renaissance tapestry fragment

The first application of UV–vis in situ fluorimetry to original art-works was experimented on a fragment from Renaissance tapestry (Fig. 13), supplied by the restorer Enrico Berardi.

The fluorescence spectra taken from the front and back sides are shown in Fig. 14. The spectrum monitored on the back side showed two distinct bands, one at 465 nm and one at 620–630 nm, whereas, on the front side, the latter was reduced to a shoulder. This indicates that the 465 nm band corresponds to the textile emission, while the dye, whose emission is at longer wavelengths, was degraded on the light-exposed front surface.

The comparison of the emission taken from the back of the original textile with a laboratory sample dyed with orcein is shown in Fig. 15. The correspondence is very good and allows the colour to be assigned to orcein, and the textile to wool, as also argued by visual analysis.

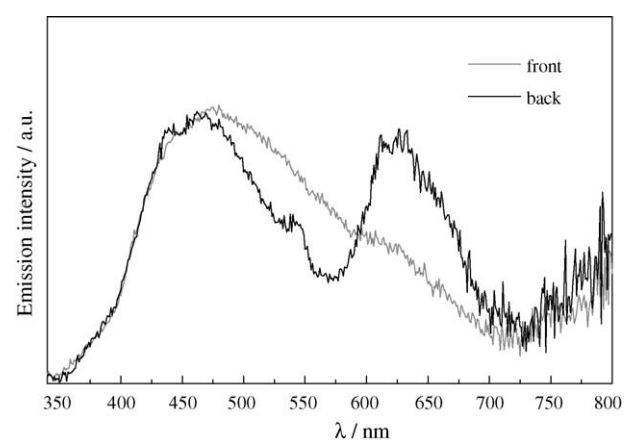


Fig. 14. Emission spectra (λ_{exc} = 254 nm) monitored using the laboratory spectrofluorimeter.

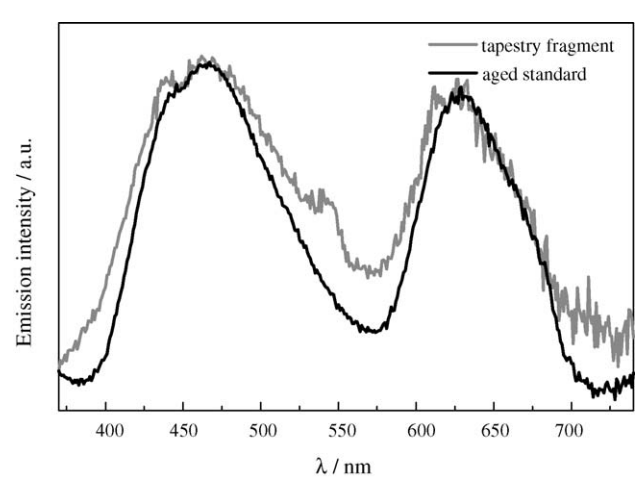


Fig. 15. Comparison of the emission from the original tapestry fragment (grey) and that of a standard dyed with orcein (black) and aged for 100 h. The spectra are normalised.

4. Conclusions

The results of this work show the potentiality of the spectrophotometric and mainly fluorimetric technique for identifying colourants on textiles, even though the absorption and fluorescence bands are rather broad when recorded on the textile surface. Spectrophotometry and fluorimetry are reliable, sensitive, capable of being utilized on a micro scale and are amenable to in situ application. However, they cannot be considered exhaustive by themselves to investigate artistic textiles. A piece of information can be obtained, but, as most of techniques, they need to be integrated by complementary experimental approaches.

This study, has confirmed that fluorescence spectroscopy is a non-invasive technique capable to distinguish dyes on surfaces and also to identify them if suitable standards are available. In fact, reflectance fluorescence patterns of organic dyes are markedly dependent on the microenvironment. In order to achieve a reliable identification of dyes on artistic objects a wide database of dyes on different substrates is necessary. Therefore, for investigating artistic textiles, standards of threads dyed using various dyes and different dyeing techniques need to be prepared.

The sensitivity and good spatial resolution of fluorescence, coupled with in situ colourimetry, allows chromatic variations consequent to photodegradation reactions to be monitored on line. The results on ageing here reported indicate that both the thread and dye are photodegradable. The extent of degradation is strongly influenced by the nature of the medium and is particularly favoured when the dye is linked to silk. The spectral changes induced by artificial photo-ageing on textiles indicate the transformations that might occur indoors in a museum. They may also be used to interpret this effect in terms of the interactions of the dye with its environment.

Acknowledgements

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References

[1] P. Novotna, V. Pacakova, Z. Bosakova, K. Stulýk, J. Chromatogr. A 863 (1999) 235–241.
[2] D.M. Grim, J. Allison, Int. J. Mass Spectrom. 222 (2003) 85–99.
[3] D. Fabbri, G. Chiavari, H. Ling, J. Anal. Appl. Pyrol. 56 (2000) 167–178.
[4] F.W. Billmeyer Jr., R. Kumar, M. Saltzman, J. Chem. Ed. 58 (1981) 307–313.
[5] E. Renè de la Rie, Stud. Conserv. 27 (1982) 1–7;
E. Renè de la Rie, Stud. Conserv. 27 (1982) 65–69;
E. Renè de la Rie, Stud. Conserv. 27 (1982) 102–108.
[6] A. Wallert, Stud. Conserv. 31 (1986) 145–155.
[7] M. Clarke, Proceedings of the Sixth International Conference on Non-destructive Testing and Microanalysis for the Diagnostics and Conservation of the Cultural and Environmental Heritage (Art '99), vol. II, Rome, 1999, pp. 1421–1436.
[8] M. Clarke, Proceedings of the Seventh International Conference on Non-destructive Testing and Microanalysis for the Diagnostics and Conservation of the Cultural and Environmental Heritage (Art 2002), Antwerp, 2002.
[9] S. Shimoyama, Y. Noda, Dyes Hist. Archaeol. 13 (1994) 14–26.
[10] D. Cardon, Le monde des teintures naturelles, Belin, Paris, 2003, pp. 371–417.
[11] T.M. Brown, C.J. Cooksey, A.T. Dronsfield, Educ. Chem. (2001) 151–153.
[12] G.W. Taylor, P. Walton, Dyes Hist. Archaeol. Text. 2 (1983) 14;
G.W. Taylor, Chem. Brit. (1990) 1155–1158.
[13] H. Robiquet, Ann. Chim. Phys. 42 (1829) 236–257;
H. Robiquet, Ann. Chim. Phys. 58 (1835) 320–335.
[14] A. Kok, Lichenologist 3 (1966) 248–272.
[15] R.M. Issa, S.M. Abu-El-Wafa, J. Chin. Chem. Soc. 31 (1984) 41–48.
[16] H. Schweppe, Handbuch der Naturfarbstoffe Vorkommen—Verwendung—Nachweis, Ecomed, Landsberg, 1993, pp. 517–534.
[17] H. Musso, Naturwissenschaften 42 (1955) 513;
H. Musso, Chem. Ber. 89 (1956) 1659–1673;
H. Musso, H. Beecken, Chem. Ber. 90 (1957) 1808–1814;
H. Musso, H.-G. Matthies, H. Beecken, H. Kramer, Angew. Chem. 69 (1957) 178.
[18] H. Musso, Planta Medica 8 (1960) 431A–446A;
H. Musso, H. Beecken, Chem. Ber. 94 (1961) 585–600;
H. Musso, S.-P. Mannsfeld, Chem. Ber. 94 (1961) 2585–2589.
[19] H. Musso, H. Beecken, Chem. Ber. 90 (1957) 2190–2194;
H. Musso, H. Kramer, Chem. Ber. 91 (1958) 2001–2016;
H. Musso, H.-G. Matthies, Chem. Ber. 90 (1957) 1814–1827.
[20] T. Philipova, C. Ivanova, Y. Kamdzhilova, M.T. Molina, Dyes Pigments 53 (2002) 219–227.
[21] C. Miliani, A. Romani, G. Favaro, J. Phys. Org. Chem. 13 (2000) 141–150.
[22] G. Favaro, C. Miliani, A. Romani, M. Vagnini, J. Chem. Soc. Perkin Trans. 2 (2002) 192–197.
[23] C. Miliani, A. Romani, G. Favaro, Spectrochim. Acta Part A 54 (1998) 581–588.
[24] P. Cox Crews, J. Am. Inst. Conserv. 21 (1982) 43–58.
[25] R.S. Davidson, J. Photochem. Photobiol B: Biol. 33 (1996) 3–25.
[26] K. Schäfer, J. Soc. Dyers Col. 107 (1991) 206–211.
[27] G. Kortuem, Reflectance Spectroscopy, Springer-Verlag, Berlin, 1969.
[28] M.R. Massafra, E. Selli, S. Salsa, B. Marcandalli, Dyes Pigments 40 (1999) 171–180.